When Audit Quality Fails to Predict Downstream Utility: Synthetic-Data Selection for Low-Resource Classification[*]

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Abstract

Quality-aware synthetic-data selection assumes that examples rated highly by an LLM judge will help downstream learning. We test this assumption in a controlled replay across two low-resource African-language classification tasks, four languages, and seven selectors. Six selectors retain 64 examples per task-language cell. CoSDA-V2 yields the highest judged label correctness (0.904 versus 0.767 for naive) and the lowest measured shortcut score, but AlpaGasus leads CoSDA-V2 on downstream Macro-F1 in both the main replay (0.202 versus 0.163) and the three-seed refit (0.193 versus 0.157).

Adding the audit's strict gate to AlpaGasus reduces Macro-F1 by 0.029, driven by one cell where only 31 of 64 candidates survive. Because the gate changes pool composition and training-set size, this comparison measures their joint effect. LESS also falls below naive when unconstrained top-k collapses label balance, but recovers with quotas. At this budget, retention constraints account for more of the observed differences than a consistent ranking advantage. In this replay, audit scores do not reliably predict downstream utility. We therefore recommend reporting both metrics for the same retained sets.

1 Introduction

Synthetic data lowers supervision costs but creates a selection problem: generators produce more candidates than learners can use, varying in correctness, redundancy, leakage, shortcuts, and coverage. Existing methods rank by judged quality, quality with complexity and diversity, validation influence, or a learned quality model (Chen et al., 2024; Liu et al., 2024; Xia et al., 2024; Wettig et al., 2024).

[*]Code, audit records, retained pools, and the claim ledger: https://github.com/lexuanbach/CoSDA.
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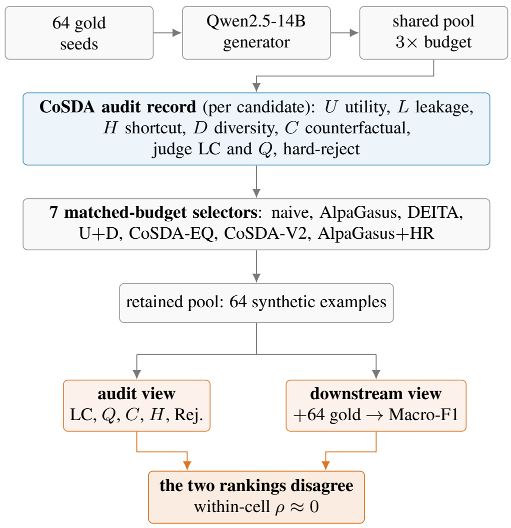

Figure 1: All selectors read the same audit records and retain 64 examples, except gate-underfilled AlpaGasus+HR. Audit scores and Macro-F1 induce different selector rankings.

Curated-data work uses hand-selected sets (Zhou et al., 2023). We test whether these signals predict utility under a matched retained-set budget in low-resource multilingual classification.

We run a controlled replay over eight task-language cells from MasakhaNEWS (Adelani et al., 2023) and AfriSenti (Muhammad et al., 2023): news topic and sentiment classification in Amharic, Hausa, Swahili, and Yoruba. All selectors read the same audit records (Figure 1), so only the selection rule changes. We compare naive with AlpaGasus, DEITA, a local utility-plus-diversity control, CoSDA-EQ, CoSDA-V2, and AlpaGasus+HR. §4.4 implements LESS itself (Xia et al., 2024). We restrict our claims to this controlled replay.

What we found. Every quality-aware selector outperforms naive on LC, Q, and residual hard-reject rate. CoSDA-V2 leads LC at 0.904 versus 0.767 for naive and has the lowest shortcut score, whereas AlpaGasus leads Macro-F1. Mean within-cell LC-to-Macro-F1 Spearman is ρ=0.04 over the five main selectors and -0.002 over seven selectors and three seeds. Two of three judge-free channels also rank selectors differently from Macro-F1, so the mismatch is not specific to the main judge.

What our analyses suggest. Adding the strict gate to AlpaGasus lowers mean Macro-F1 by 0.029 over three seeds. Amharic news dominates this change: only 31 of 64 candidates survive and Macro-F1 falls by 0.252, while the net effect across the other seven cells is $+0.003$. LESS likewise finishes below naive when unconstrained top-k puts 77% of its pool in one label, but reaches 0.278 under quotas. Both comparisons implicate retention constraints, although neither separates ranking, composition, and training-set size (§5.1).

Contributions. We make three contributions. First, we define five per-example channels for auditing a retained pool before training, spanning utility, leakage, shortcut, diversity, and counterfactual consistency, with a fixed hard-reject decision (§2). Second, we evaluate seven selectors in a matched 8-cell replay and test whether audit and downstream rankings agree across cells, seeds, backbones, and judges (§3 to §5). Third, we release the per-candidate audit records, the retained pools, and a claim ledger that traces each reported aggregate to its source (§A).

2 CoSDA: A Counterfactual Audit

We use CoSDA to make a selected pool inspectable before it reaches training. Every selector we compare reads the same per-candidate record, so any difference between their pools is interpretable against a shared trace.

Setup. For task t, language ℓ, and gold budget b, let $G_{t,\ell,b}$ be the gold seed set. The generator returns a candidate pool $X_{t,\ell,b}$ with multiplier m, and each x_i carries a proposed label y_i, generator metadata, source seed IDs, and a counterfactual pair ID. Our replay uses Qwen2.5-14B-Instruct (Qwen Team, 2024) at p=0.95 and T=0.8 as generator, Qwen2.5-32B-Instruct-AWQ as judge, and XLM-RoBERTa base (Conneau et al., 2020) as classifier. The generation prompt fixes task, language, label schema, seed examples, output format, and a forbidden-behaviour clause forbidding copied seed or held-out text and literal English-template translations. Appendix B reproduces the template, whose length hint branches by task.

Counterfactual pair construction. CoSDA builds task-specific minimal pairs $c_i = (x_i, x'_i)$ to ask whether a candidate behaves like a labelled example. The generator flips the topic or sentiment cue while holding language, register, and non-label content fixed, and the expected flipped label y'_i feeds C_i. The C_i gate pass rate, $C_i \geq 0.6$, varies by language at 33.4% for Amharic, 52.9% Hausa, 55.2% Swahili, and 58.9% Yoruba. Amharic therefore enters selection with a much smaller usable counterfactual pool, and that asymmetry matters in §5.1. The validator is deliberately task-specific, so extending CoSDA to other task families means defining the corresponding transformation.

Audit channels. Each candidate receives five normalised scores in $[0, 1]$:

$$U_i = \tfrac{1}{3}(\sigma_{\text{gen}} + a_{\text{teach}} + v_{\text{nbr}}), \tag{1}$$
$$L_i = \max\{E_i, J_i, B_i\}, \tag{2}$$
$$H_i = \textstyle\sum_k w_k \phi_k(x_i), \tag{3}$$
$$D_i = 1 - \max_{j\in G} \cos(\mathbf{e}_i, \mathbf{e}_j), \tag{4}$$
$$C_i = v_{\text{cf}}\big(0.75 + 0.25\, p_\theta(y'_i \mid x'_i)\big). \tag{5}$$

We define *utility* U_i as the average of the judge's quality score (or generator confidence where no judge ran), the confidence a LaBSE-centroid teacher fit on $G_{t,\ell,b}$ assigns to y_i, and a k=5 gold-neighbour label vote in LaBSE space (Feng et al., 2022). We use a centroid teacher because fine-tuning one on 64 examples is unstable. We measure observable *leakage* L_i as the maximum of an exact 10-gram overlap indicator E_i against the dev and test union, the largest 5-gram shingle Jaccard J_i against any held-out item (Broder, 1997), and B_i, a LaBSE cosine rescaled so only similarity above 0.85 contributes.

Our *shortcut* score H_i is a bounded artifact-risk score, a fixed weighted sum $\sum_k w_k \phi_k$ over label-word hits, template markers, length atypicality, an entity proxy, and punctuation ratio. It is a heuristic rather than a trained probe, so it flags surface regularities rather than certifying their absence. We compute *diversity* D_i as one minus the maximum LaBSE cosine of x_i against the gold seeds, once and not updated as selection proceeds. We define *counterfactual consistency* C_i by scaling the judge's validity rating v_{cf} with a soft teacher term that rewards probability mass on the expected flipped label y'_i. We use a soft term rather than a hard indicator because the teacher is noisy at this budget.

The same record stores the two judge fields quality-only selectors consume, label correctness and quality. Label correctness sits outside S_i, while quality enters it through U_i, and we keep both so judge-facing quality and downstream utility can be compared on identical pools. Every audit metric in §4 is averaged over each selector's *retained* pool, 64 examples except for the two underfilled AlpaGasus+HR cells, so audit channels and Macro-F1 describe the same training set. A candidate violates the gate when it crosses a fixed leakage, shortcut, or counterfactual-consistency threshold, so the rate we report is the *residual* violation rate inside it.

The hard-reject gate and the soft score. The audit defines a strict gate,

$$\mathrm{reject}(x_i) = \mathbf{1}[L_i{>}\tau_L \vee C_i{<}\tau_C \vee H_i{>}\tau_H],$$

with $\tau_L{=}0.15$, $\tau_C{=}0.60$, and $\tau_H{=}0.70$ fixed before the replay, alongside a five-term soft score

$$S_i = \alpha_U U_i + \alpha_C C_i + \alpha_D D_i - \alpha_L L_i - \alpha_H H_i,$$

with $\alpha_U{=}1.0$, $\alpha_C{=}0.75$, $\alpha_D{=}0.50$, $\alpha_L{=}1.0$, $\alpha_H{=}0.75$. Both belong to the audit record, and our two selectors differ in ranking rule and in how much authority each gives the gate (§5.1).

CoSDA-EQ and CoSDA-V2. CoSDA-EQ treats rejection as a preference rather than a veto, ranking gate-passing candidates by S_i and backfilling with the rejected ones that score highest on the same S_i.

CoSDA-V2 drops hard rejection altogether. Every candidate stays eligible, and the thresholds re-enter only as graded penalties and small bonuses,

$$S_i^{\mathrm{V2}} = P_i + A_i + 0.25\,\beta_i^{\mathrm{s}} + 0.10\,\beta_i^{\mathrm{r}} - R_i,$$

with a judge block $P_i = 1.25\,\mathrm{LC}_i + 0.85\,Q_i + 0.45\,f_i + 0.35\,\kappa_i + 0.35\,U_i$ over judged label correctness, quality, fluency, and cultural plausibility, an audit block $A_i = 0.65\,C_i + 0.45\,D_i$, and a risk penalty

$$R_i = 1.4L_i + H_i + 1.1\lambda_i + 0.55(\tau_C{-}C_i)_+ + 0.45(L_i{-}\tau_L)_+ + 0.35(H_i{-}\tau_H)_+,$$

where λ_i is the judge's leakage suspicion and $(\cdot)_+$ the positive part, and $\beta_i^{\mathrm{s}}, \beta_i^{\mathrm{r}}$ mark candidates clearing the strict gate and a relaxed one ($\tau_C{\to}0.45$). We built V2 this way because the earlier hard-gated variants retained high-scoring candidates but left too little coverage in the cells with the smallest usable pools. Since V2 never removes a violator, its residual hard-reject rate can stay above zero.

Each selector but naive retains its top k under per-label quotas rounded to the gold prior and a cap of three candidates per source seed, with overflow deferred and used as backfill. We calibrated τ_L and the α weights on a held-out Setswana split outside the replay and fixed τ_H and τ_C by protocol. The S_i^{V2} coefficients were set by hand before the replay and were not tuned on the reported cells. We do not release the calibration run. We therefore treat these values as fixed protocol choices, not as reproducible derivations (see Limitations).

3 Experimental Setup

Tasks and languages. We evaluate an 8-cell replay over MasakhaNEWS news topic classification (Adelani et al., 2023) and AfriSenti sentiment classification (Muhammad et al., 2023), in Amharic, Hausa, Swahili, and Yoruba. Macro-F1 is the metric in every cell, and Appendix Table 8 gives splits and provenance. We treat each cell independently and aggregate after inspecting per-cell outcomes, since three of our eight turn out near-degenerate.

Low-resource protocol. Every cell gives us $b{=}64$ gold examples and a generation multiplier $m{=}3$, so each selector chooses 64 synthetic examples from roughly $3b$ candidates. Our main replay is deterministic at seed 13, which means every selector draws from the same trace-backed candidate pool in each cell. That makes the comparison a within-pool test of selection criteria rather than of independently sampled generations.

Generation and scoring. All selectors read a shared candidate pool per cell, scored per §2 with generator and judge quality signals, centroid-teacher agreement, LaBSE-based diversity and leakage checks (Feng et al., 2022), shingle-overlap near-duplicate checks (Broder, 1997), shortcut features, and counterfactual scores.

Selectors. Seven selectors read that pool. All but naive apply the same label-quota and source-seed constraints. Six retain 64 examples in every cell. AlpaGasus+HR retains 31 and 50 in two cells and 64 elsewhere. **(1) Naive** shuffles the pool at the cell seed and keeps 64 candidates, using no audit signal and, unlike the others, no balance constraint.

(2) AlpaGasus-style (Chen et al., 2024) keeps the top 64 by judged quality. **(3) DEITA-style** (Liu et al., 2024) keeps the top 64 by judged quality, embedding diversity, and a length-based complexity proxy. **(4) U+D control** ranks by U_i+$0.25D_i$, a gold-fit proxy. §4.4 additionally implements LESS itself (Xia et al., 2024) and compares the two. AlpaGasus and DEITA are budget-matched reimplementations of the published *criteria*. The U+D row is a local control, not a reimplementation of LESS. **(5) CoSDA-EQ** ranks gate-passing candidates by S_i and backfills with the least risky rejected ones. **(6) CoSDA-V2** is our budget-preserving reranker, which applies no gate and takes the top 64 by S_i^{V2} (§2). **(7) AlpaGasus+HR** is the ablation selector for §5.1. It applies the audit's strict gate, the one CoSDA-V2 declines to use, and ranks the survivors by AlpaGasus's score. We also fine-tune a Gold-only reference on the gold examples alone and report it in Appendix C, keeping it out of the main comparison since it has no synthetic pool to audit.

Models and metrics. We fine-tune the deterministic Hugging Face classification path used in the CoSDA replay and report Macro-F1. In every cell we train on the union of the 64 gold seeds and the selector's retained synthetic examples, giving 128 rows for every selector except AlpaGasus+HR, whose gate leaves only 31 and 50 candidates in two cells. The gold seeds are identical across selectors, so the retained synthetic set is the only thing that varies. We compute audit metrics over each selector's retained examples, namely LC, Q, counterfactual consistency C, leakage L, shortcut H, and residual hard-reject rate. The main replay uses XLM-RoBERTa base (Conneau et al., 2020), and §4.4 repeats the fit with AfroXLMR base (Alabi et al., 2022) on the same pools.

Extended replay. Our new selectors, the multi-seed re-fit, and the second backbone all reuse the seed-13 retained pools, so audit channels cannot move and only the downstream fit is recomputed. The extension comes to 224 further fine-tunes, covering 7 selectors over 8 cells at seeds 13/21/42 on XLM-R, plus 7×8 on AfroXLMR at seed 13. It ran on different hardware from the original replay, and per-selector Macro-F1 moves by up to 0.048 against Table 1, enough to reorder the selectors at the shared seed. We use that difference only to document implementation sensitivity, and we report each pipeline separately rather than pooling their absolute Macro-F1 values.

Selector	LC↑	Q↑	C ↑	H ↓	Rej.↓	F1↑
Naive	0.767	0.662	0.587	0.066	0.486	0.167
AlpaGasus	0.890	**0.749**	0.653	0.065	0.246	**0.202**
DEITA	0.869	0.737	0.637	0.070	0.299	0.128
CoSDA-EQ	0.836	0.718	**0.686**	0.058	**0.109**	0.133
CoSDA-V2	**0.904**	0.746	0.674	**0.057**	0.162	0.163
Spearman ρ (LC-rank vs. F1-rank)						+0.10

Table 1: Audit channels and downstream utility on the same retained pools, averaged over 8 cells (seed 13). Rej. is the residual hard-reject rate. Every quality-aware selector beats naive on LC, Q, C, and Rej., but only AlpaGasus also wins downstream.

4 Results: The Quality-to-Utility Gap

We compare selectors using two judge-derived quantities, each averaged over that selector's retained pool: *label correctness* (LC), the judge's confidence that y_i is right for x_i, and *quality* (Q), its holistic rating of suitability for fine-tuning. Ranking selectors by these audit scores and then by Macro-F1 gives two orderings that disagree, within cells and across three seeds as well as in the main replay, though the size of the gap depends on backbone and judge.

4.1 The gap

Every quality-aware selector in Table 1 improves on naive on LC, Q, C, and residual hard-reject, and all but DEITA also on shortcut H, even though the individual methods do not optimise every channel directly. Our hard-reject rate is the *residual* rate among retained examples, so it describes what a selector hands to training (§2). Our downstream evaluation yields a different ordering. AlpaGasus is the only selector that improves on naive Macro-F1, by 0.036, while DEITA, CoSDA-EQ, and CoSDA-V2 fall below naive despite carrying better audit profiles, and CoSDA-EQ alone holds the same rank in the audit and downstream orderings. We measure the selector-level Spearman between mean LC and mean Macro-F1 at ρ=0.10, and between hard-reject and Macro-F1 at ρ=0.20, but with five selectors we treat these as descriptive.

We next test whether the mismatch depends on judge-derived scores. Three channels use no judge field. Oriented so that higher is better, diversity is inversely ranked with Macro-F1 ($\rho = -1.00$) and leakage likewise (-0.40), while shortcut is weakly aligned ($+0.20$). These are descriptive checks rather than evidence that every audit channel di-

Attribute	AlpaGasus	CoSDA-V2	Δ
Macro-F1↑	**0.202**	0.163	+0.039
LC↑	0.890	**0.904**	−0.014
Q↑	**0.749**	0.746	+0.003
Hard reject↓	0.246	**0.162**	+0.084
C↑	0.653	**0.674**	−0.021
Per-cell wins↑	3	3	(2 ties)

Table 2: Comparison of the audit and downstream leaders. CoSDA-V2 leads three of four audit rows, while AlpaGasus has higher Macro-F1. Bold marks the better value. Per-cell wins are counted on unrounded Macro-F1: sent./yor is an AlpaGasus win of 0.0006 that rounds to a tie in Appendix Table 11.

verges from utility, and we do not combine them because their scales differ by an order of magnitude. Counterfactual consistency is judge-assisted, since C_i scales the judge's validity rating, and the residual reject rate inherits that. The mismatch therefore appears in channels that read no judge field.

4.2 Audit leader vs. downstream leader

Our comparison of the two leaders isolates the practical mismatch (Table 2). CoSDA-V2 scores better on three of the four displayed audit measures, namely LC, residual hard-reject rate, and C, trailing AlpaGasus by 0.003 on Q, yet AlpaGasus trains the stronger classifier by 0.039 Macro-F1. Because each selector wins three cells and two tie, that aggregate difference reflects the size of cell-level effects rather than a uniformly better selector.

Within-cell ranks. We use within-cell correlations because every selector then faces the same pool and test set, which an aggregate across eight heterogeneous cells cannot guarantee (Figure 2). Their signs split evenly, four positive including Swahili news at +0.87 and Hausa sentiment at +0.90 and four negative, with mean +0.04 and median 0.00. We therefore read the alignment as heterogeneous and unstable rather than as a uniform negative relation, and we treat this as our primary statistic.

4.3 Robustness

Cell heterogeneity. Our sensitivity analyses expose two distinct forms of heterogeneity. First, the near-zero −0.004 mean for CoSDA-V2 against naive rests on one cell: two are strongly positive, Swahili news at +0.129 and Hausa sentiment at +0.206, and dropping Hausa sentiment alone moves the mean to −0.034, in line with DEITA and CoSDA-EQ (Appendix Table 11, Table 13, Figure 5). We therefore do not read the aggregate as a

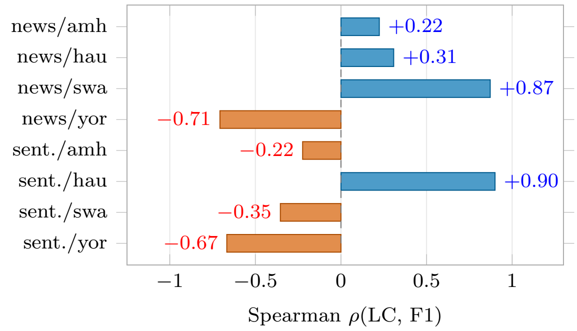

Figure 2: Within-cell Spearman between judged label correctness and Macro-F1, computed across selectors inside each cell. Four cells are positive and four negative, with mean +0.04.

stable V2 advantage. Second, three cells are near-degenerate, with the five selectors spanning under 0.07 Macro-F1. After we remove them the LC-to-F1 Spearman climbs to +0.50, yet AlpaGasus keeps a wider lead over CoSDA-V2 (Appendix C). The informative subset therefore shows closer rank alignment without eliminating the disagreement between the two leaders.

Uncertainty. Resampling the 8 cells gives 95% intervals on the mean delta over naive of $[-0.079, +0.078]$ for CoSDA-V2 and $[-0.022, +0.093]$ for AlpaGasus. On the five non-degenerate cells, $[-0.114, +0.133]$ at mean +0.007 and $[-0.015, +0.137]$ at mean +0.070. Both intervals contain zero, so we do not conclude that CoSDA-V2 or AlpaGasus improves on naive Macro-F1. We still observe a difference between the audit and downstream rank orders, but these intervals bound downstream improvement and do not quantify uncertainty in that difference.

Audit improvements. CoSDA-V2 lifts LC by 17.9% relative to naive, and CoSDA-EQ cuts residual hard-reject by 77.5% relative while raising C from 0.587 to 0.686. These are measured properties under the defined audit channels. §5 takes up how pools with higher audit scores can still finish behind on Macro-F1.

4.4 Extended replay

Our main replay fixes one seed, backbone, judge, and five selectors. We relaxed each in turn on the seed-13 pools, so audit columns cannot move and only the downstream fit or judge changes.

Three seeds. Re-fitting all seven selectors at seeds 13, 21, and 42 (Table 3) shows that the downstream ordering is seed-sensitive even though

Selector	LC↑	XLM-R F1↑ (3 seeds)	AfroXLMR↑ (seed 13)
CoSDA-V2	**0.904**	0.157 ± 0.018	0.344
AlpaGasus	0.890	$\mathbf{0.193 \pm 0.024}$	0.321
DEITA	0.869	0.174 ± 0.011	0.325
U+D control	0.849	0.178 ± 0.010	**0.348**
AlpaGasus+HR	0.845	0.164 ± 0.008	0.339
CoSDA-EQ	0.836	0.167 ± 0.010	0.341
Naive	0.767	0.171 ± 0.038	0.266

Table 3: Extended replay on the seed-13 pools, sorted by audit LC. CoSDA-V2 ranks last in mean XLM-R Macro-F1 across three seeds but second under AfroXLMR at seed 13. Values after $\pm$ are the seed standard deviation.

Selector	F1↑	Retained pool
U+D control	0.297	label quotas
AlpaGasus	0.288	label quotas
LESS + quotas	0.278	label quotas
Naive	0.205	unconstrained
LESS (as specified)	0.138	unconstrained top-k

Table 4: Real LESS against the selectors it competes with, eight cells at seed 13 under one identical training loop. Absolute values are not comparable to Table 1, which used a different pipeline.

the retained pools and their audit scores are fixed. CoSDA-V2 moves from second of seven at seed 13 to sixth at seed 21 and last at seed 42, and the selector-level LC-to-F1 Spearman runs $+0.46$, $+0.43$, and -0.29 across the same seeds. Our within-cell statistic also changes sign, over a narrower range, at $+0.006$, $+0.233$, and -0.246. Its seed-13 value is $+0.006$ here and $+0.04$ in Figure 2 because this pipeline ranks seven selectors rather than five and refits them on different hardware, so the two are not interchangeable. We therefore treat no single seed as a stable estimate of audit-utility alignment. The means keep the inversion, with CoSDA-V2 worst on Macro-F1 at 0.157 and AlpaGasus best at 0.193.

A validation-aware selector. A selector that optimises for downstream fit directly is the most relevant comparison for our claim, so we implemented LESS (Xia et al., 2024) on these pools: LoRA warmup with AdamW over four checkpoints, per-example gradients, Adam preconditioning in full parameter space, plain validation gradients, and the checkpoint-weighted cosine of its Definition 3.1, computed exactly rather than through the paper's random projection. Table 4 reports it beside its competitors under one identical training loop.

As specified, LESS finishes last, below naive, and our comparisons implicate its selection rule rather than its scores. LESS ranks candidates and then takes a plain global top-k with no balance constraint, which at a 64-example budget puts 77% of the retained pool in one label on average and 98% in all four sentiment cells, while Macro-F1 averages over classes. The ranking itself is informative: keeping the bottom 64 rather than the top 64 drops Swahili news from 0.181 to 0.012. Given the label quotas the other selectors receive, LESS reaches 0.278, near AlpaGasus at 0.288. We do not read this as evidence against validation-aware scoring. Under the 64-example protocol, enforcing label quotas moves LESS by 0.140 Macro-F1, whereas changing the ranking under the same quotas moves it by at most 0.019. In this replay, set composition dominates the difference among the ranking rules we test.

A second backbone. Re-fitting the same pools with AfroXLMR base (Alabi et al., 2022), an in-domain African-language encoder, changes the relation between learner and ranking (Appendix Figure 4). Against the extended pipeline's own seed-13 XLM-R fits every selector gains, least of all naive at $+0.118$, the within-cell Spearman rises from $+0.006$ to $+0.21$, and CoSDA-V2 is second at 0.344 against 0.321 for AlpaGasus, while naive stays worst at 0.266. Because we evaluate AfroXLMR at one seed, we read the narrower gap as a backbone interaction and do not attribute it to language representation. The two rankings align better under the higher-scoring backbone, so the gap we measure depends on the learner as well as on the retained pool.

Three judges. We re-scored label correctness on the same pools with two additional judges. Qwen2.5-14B-Instruct, our generator, is a smaller model from the same family as our primary judge, while Phi-3.5-mini-instruct (Abdin et al., 2024) changes both family and size. Each scored 1,523 candidate records, of which 1,521 and 1,502 returned a usable score. Table 5 reports mean label correctness for all seven selectors under the three judges, with the cross-judge Pearson correlation in the final row.

Our additional judges (Table 5) separate coarse screening from fine ranking. Qwen-14B keeps every quality-aware selector above naive and correlates with our original judge at Pearson $+0.73$. Phi-3.5 weakens that separation, correlating only $+0.29$ and placing three of the six below naive. More importantly, neither judge reproduces the ordering

Selector	Qwen-32B	Qwen-14B	Phi-3.5
CoSDA-V2	**0.904**	0.556	0.417
AlpaGasus	0.890	0.538	0.425
DEITA	0.869	0.495	0.411
U+D control	0.849	**0.557**	**0.426**
AlpaGasus+HR	0.845	0.518	0.403
CoSDA-EQ	0.836	0.538	0.406
Naive	0.767	0.458	0.414
Pearson vs. Qwen-32B		+0.73	+0.29

Table 5: Label correctness under three judges on the same pools. Qwen-14B preserves the quality-aware-vs-naive separation. Phi-3.5 does not. The ordering among quality-aware selectors is judge-sensitive throughout.

among the quality-aware selectors: the U+D control leads under both replacements, at 0.557 and 0.426, and CoSDA-V2 falls to third under Phi-3.5. We therefore read LC as a judge-specific screening measure rather than a stable selector ranking.

Does the judge track human labels? We checked the judge against human-annotated benchmark labels. We drew 320 human-labelled items, 160 per task, and scored each twice, once correctly labelled and once flipped, giving 640 presentations. It accepts a correct label 50.3% of the time and a flipped one 14.1% at 68.1% agreement, a margin of 36.25 points before rounding, which the released record reports as 36.3. The margin is far wider on news topic, 56.9% against 6.2%, than on sentiment, 43.8% against 21.9%. We use this only as evidence that the judge separates correct from flipped labels under one prompt, not as validation of the generated text. We do not release the raw judgments or the acceptance rule.

5 Interpreting the Audit-Utility Mismatch

We next examine why our audit ranking and our downstream ranking diverge. Per-example audit scores and set-level training utility measure different properties of the same pool, and the trace data do not identify a single mechanism. They support three bounded interpretations, which we treat as explanations to be tested rather than as a causal decomposition.

Residual risk and gate underfill. We first ask whether residual audit risk is associated with downstream loss. CoSDA-V2 rejects nothing, so its residual violation rate cannot measure underfill. That rate is nevertheless higher where V2 loses to naive, averaging 0.23 across those five cells against 0.08 in the two where it wins, and correlating -0.56 with the V2-minus-naive delta (p=0.15, two-sided permutation). With eight cells we read this as a descriptive association. §5.1 then tests a distinct hypothesis, that applying the strict gate reduces the support available to the classifier when too few candidates survive.

Diversity does not explain the AlpaGasus lead. Quality ranking might preserve more variety, though D_i measures distance from gold rather than within-pool coverage. AlpaGasus scores D=0.590, the lowest of the five selectors, below naive at 0.599 and below both CoSDA variants at 0.602 and 0.604, while DEITA, which optimises for diversity, scores highest at 0.649 and still finishes last downstream (Appendix Table 10). Our data therefore rule out mean distance from gold as the explanation, but they do not identify why AlpaGasus leads. One possibility is decision-boundary support carried by examples that score well on judged quality but fail the audit's counterfactual gate.

Limited resolution in judge scores. We observe a compressed range in both judge channels: every selector scores at least 0.76 on LC with the field bunched near 0.90, while Q lies in $[0.66, 0.75]$, and the two rank selectors almost identically at $+0.90$. We hypothesise that this compression limits their ability to distinguish selector differences at the scale Macro-F1 responds to. Our multi-judge check (§4.4) supports the broader claim of ranking sensitivity, since the fine ordering reshuffles across judges while the coarse separation is steadier, but it does not directly measure saturation.

What our evidence distinguishes. The three interpretations are non-exclusive. Only the first receives a direct intervention (§5.1), while the other two rest on trace evidence.

5.1 Ablation

The underfill hypothesis holds that audit-strict filtering removes support the classifier needs. CoSDA-V2 itself never hard-rejects (§2), so we cannot test this by removing a step from V2. We test the gate directly instead. AlpaGasus+HR applies the audit's strict gate and ranks the survivors by AlpaGasus's judged-quality score, holding the ranking rule fixed at the downstream leader's. It measures what hard gating costs a strong selector, which is precisely the choice CoSDA-V2 declines to make.

Table 6 and Figure 3 localise the cost. Adding

Cell	HR pool	AG↑	AG+HR↑	Δ
news/amh	**31**	0.387	0.135	**−0.252**
news/hau	**50**	0.114	0.140	+0.026
news/swa	81	0.145	0.078	−0.067
news/yor	87	0.067	0.133	+0.066
sent./amh	93	0.220	0.153	−0.067
sent./hau	153	0.268	0.331	+0.063
sent./swa	131	0.155	0.155	+0.000
sent./yor	138	0.184	0.184	+0.000
mean		0.193	0.164	−0.029

Table 6: Applying the audit's strict gate to AlpaGasus's ranking, averaged over seeds $13/21/42$. HR pool counts candidates surviving the gate against a budget of 64. The mean is dominated by news/amh.

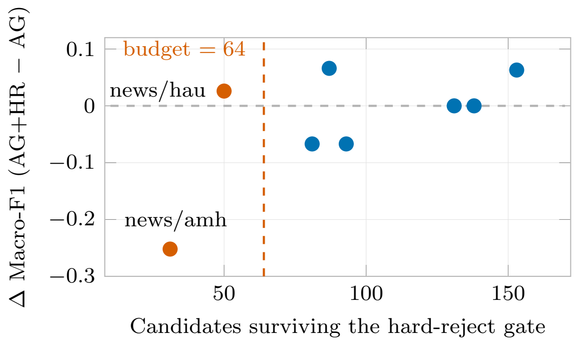

Figure 3: Effect of adding the audit's strict gate to AlpaGasus, against how many candidates survive it. The mean loss is dominated by the one cell where the surviving pool falls far below the 64-example budget.

the gate costs 0.029 mean Macro-F1 over three seeds, and one cell dominates that average: in Amharic news the gate admits only 31 of the 64 candidates the budget calls for, and Macro-F1 falls by 0.252. Amharic also has the lowest C_i pass rate of our four languages, 33.4% against 52.9% to 58.9% elsewhere, so the gate strips its pool hardest in the cell where the loss appears. Milder starvation does not reproduce it, since Hausa news retains 50 of 64 and still gains 0.026.

Our ablation supports one narrow conclusion: in this replay the measured cost concentrates in the single cell with severe underfill. Across the six cells with enough surviving candidates its mean effect is -0.001, with cell-level effects between -0.067 and $+0.066$ that cancel rather than vanish, and sent./swa, whose pool overlaps AlpaGasus's at $63/64$, does not move in any seed. Because the contrast adds the gate while holding the ranking fixed, it is not evidence that the counterfactual and shortcut channels order candidates badly. We cannot tell whether the Amharic-news loss comes from the reduced candidate count or from the composition of the surviving pool, since both change together, and a count-matched AlpaGasus control would separate them.

5.2 Cell-level case studies

Our eight cells group three ways (Appendix Table 11). Two are positive, Swahili news and Hausa sentiment, where V2's residual rate is low at 0.16 and 0.00. Three are negative, and the largest loss carries the largest residual rate at 0.609, though the relation is not monotone, since Hausa news is the smallest loss and carries the second-largest rate at 0.406. Three are near-degenerate. This grouping is consistent with an association between residual risk and downstream loss, but eight cells do not resolve it, so we use it to motivate benchmark design rather than to explain the deltas. Future evaluations should vary candidate pass rate, label balance, and baseline task difficulty independently, so that audit stress can be separated from learner difficulty.

5.3 Implications for selector evaluation

We draw three practical implications from our replay. First, we recommend reporting audit and downstream metrics on identical retained sets, because their rankings can diverge even when generation is held fixed. Second, we treat judged label correctness as a property of one judge's screening behaviour rather than as a downstream proxy, since our selector-level Spearman is ρ=0.10 here and swings from -0.29 to $+0.46$ across seeds (§4.4). Third, we recommend making hard rejection budget-aware, since §5.1 finds the gate's cost concentrated where too few candidates survive.

When we would still prefer CoSDA-V2. CoSDA-V2 may still be preferred when its channels are objectives in themselves, such as retained leakage of L=0.003 against AlpaGasus's 0.006, the lowest retained shortcut score at H=0.057, or the only label correctness above 0.90. These describe the pool it returns, and V2 applies graded penalties rather than hard constraints, so it can still retain a violator. Our two variants differ in both scoring and gate treatment, EQ ranking by S_i with gate priority and backfill and V2 by S_i^{V2} with no gate, yet they reach similar diversity and sit inside each other's bootstrap uncertainty, so neither difference separates them downstream at this scale.

6 Related Work

Synthetic supervision and selection. Synthetic-data selection methods operate on pools produced by model-generated supervision (Wang et al., 2023; Honovich et al., 2023; Wang et al., 2022; Mishra et al., 2022; Bach et al., 2022; Longpre et al., 2023; Taori et al., 2023; Xu et al., 2024), including multilingual variants (Üstün et al., 2024; Aryabumi et al., 2024). We group them by where they locate utility. Candidate-scoring methods treat quality, complexity, or diversity as properties that transfer across learners: AlpaGasus ranks by ChatGPT quality (Chen et al., 2024), DEITA adds complexity and diversity (Liu et al., 2024), QuRating learns a quality model (Wettig et al., 2024), and related work studies self-guided scoring and small curated sets (Li et al., 2024; Cao et al., 2024; Zhou et al., 2023). Validation-aware methods instead define utility relative to a target model, as LESS does by selecting on validation influence (Xia et al., 2024). Our matched-pool replay tests whether these two views induce the same ordering under a fixed budget.

LLM judges as selection evidence. Judges scale cheaply across candidate pools, but their multilingual reliability varies with model, language, and criterion (Tan et al., 2024; Fu and Liu, 2025). High label correctness means the judge can map a candidate to its intended label, not that a small classifier will learn a better boundary from it. That distinction sharpens when a retained pool holds 64 examples, so we treat judge scores as audit channels rather than as evidence of training utility.

Utility-vs-quality misalignment. Training-dynamics and pruning studies show that example value depends on the learner and the data regime (Swayamdipta et al., 2020; Sorscher et al., 2022), which is why the methods above add criteria or condition on a target model. We instead measure the misalignment directly at the *selector* level, comparing audit and downstream rank orders on identical retained pools.

Data-centric diagnostics and counterfactual checks. We adapt three families of data-centric diagnostics to selection rather than to training. Cartography, forgetting, proxy selection, and pruning locate useful or harmful training points (Swayamdipta et al., 2020; Toneva et al., 2019; Coleman et al., 2020; Paul et al., 2021; Mindermann et al., 2022; Sorscher et al., 2022), counterfactual tests ask whether models respond to task-relevant changes rather than artifacts (Kaushik et al., 2020; Gardner et al., 2020; Ribeiro et al., 2020; Wu et al., 2021), and work on shortcuts and memorisation explains why fluent examples can still be risky (Gururangan et al., 2018; Poliak et al., 2018; McCoy et al., 2019; Geirhos et al., 2020; Jia and Liang, 2017; Dodge et al., 2021; Carlini et al., 2021; Lee et al., 2022; Kandpal et al., 2022; Maini et al., 2024). We apply them to each retained pool before asking whether the signals predict Macro-F1.

Low-resource multilingual benchmarks. Our replay uses MasakhaNEWS and AfriSenti (Adelani et al., 2023; Muhammad et al., 2023), whose four languages are written in two scripts, Ge'ez and Latin, though we do not estimate a script effect. AfroXLMR (Alabi et al., 2022) supplies the in-domain backbone for our learner-sensitivity check in §4.4.

7 Conclusion

We tested whether judged quality predicts downstream utility when selectors retain the same number of examples. In our controlled replay, it did not do so reliably. CoSDA-V2 led on several audit measures, while AlpaGasus achieved higher Macro-F1 with both the main backbone and the three-seed refit. Rankings also varied across seeds, backbones, and judges. Audit scores can therefore describe a retained pool, but they cannot replace downstream evaluation in this setting.

The gate stress test also exposed the risk of overly strict filtering. Adding the strict gate to a fixed ranking lowered mean Macro-F1 by 0.029; nearly all of this loss came from one cell where only 31 of 64 requested candidates survived. Because the gate changed both pool composition and training-set size, this result supports budget-aware filtering rather than implicating any single audit channel.

We therefore recommend reporting audit scores and downstream performance for the same retained sets. We release the per-candidate records, retained pools, and scripts so this comparison can be repeated with other generators, tasks, and budgets.

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Limitations

Scope. Our evidence is a controlled case study rather than a general result. It covers eight task-language cells in Amharic, Hausa, Swahili, and Yoruba, two classification tasks, one generator, one primary judge, and two backbones. The question we study is not specific to low-resource African languages, but our evidence is, and we interpret every result within that scope.

Cell count and statistical strength. Our replay covers 8 task-language cells. We compute the selector-level Spearman over 5 selectors in the main replay and 7 in the extension, and §4.4 shows that statistic is unstable across seeds, ranging from -0.29 to $+0.46$. That is why we report it qualitatively alongside rank orders and rest our argument on the within-cell Spearman instead. Our bootstrap CIs on V2-minus-naive and AlpaGasus-minus-naive both contain zero, so we make no claim that any selector beats naive downstream. Absolute Macro-F1 is low throughout and three cells are near-degenerate. The inversion survives when we drop those, though the LC-to-F1 rank correlation rises there too (§4.3), so the restriction cuts both ways. Every selector difference we report is therefore small in absolute terms, and we read them as rank evidence rather than as effect sizes.

Seeds and pipelines. Our main replay is a single-seed deterministic trace, and the multi-seed evidence comes from re-fitting the same retained pools at three seeds on different hardware. Per-selector Macro-F1 therefore moves by up to 0.048 against Table 1, enough to reorder the selectors at the shared seed, which is why we report the two pipelines separately rather than pooling them. Three seeds bound seed variance loosely rather than tightly. Each deterministic replay across 7×8 fine-tunes plus the judge audit costs roughly one GPU-week per seed, and that cost is what caps our seed count.

Task scope. We cover two classification tasks. Sequence labelling, intent and slot filling, and summarisation each need their own counterfactual validator (§2), and we chose to isolate the classification setting before extending the validator family. We make no claim about high-resource or generative settings.

Backbone dependence. The gap is backbone-conditioned. With XLM-RoBERTa base the audit-best selector is worst downstream, while with AfroXLMR base the within-cell Spearman rises to $+0.21$ and CoSDA-V2 recovers to second place (§4.4). We tested two backbones, and only one seed on the second, so we do not know whether the re-alignment continues with stronger encoders. Our two-backbone comparison raises the hypothesis that audit-utility divergence grows where the downstream learner is weaker on the target language. Testing it requires more backbones and seeds.

Generator dependence. All our candidates come from Qwen2.5-14B-Instruct under one nucleus-sampling configuration. Unlike the backbone and the judge, we did not vary the generator, so our finding is scoped to it. A stronger or differently tuned generator would change the composition of the pool every selector reads.

Judge dependence. Judged label correctness and quality come from a single multilingual judge prompted in English with in-language task descriptions. Our two additional judges (§4.4) show the coarse separation holds under a same-family judge but not under a much smaller one, and the fine LC ordering is unstable throughout, so the specific LC values, and the identity of the LC leader, should be read as properties of that judge and prompt. The gap finding does not rest on this channel alone, because the channels that use no judge field (D, L, H) also yield rank orders that differ from Macro-F1, and the within-cell Spearman is computed inside each cell on a shared pool. Counterfactual consistency is judge-assisted, since C_i scales the judge's validity rating. We did not run a multi-prompt sweep.

No human validation of the synthetic text. This is the most consequential limitation for an African-language study. We validated the *judge* against human-annotated benchmark labels (§4.4), where it separates correct from flipped labels by 36.25 points but agrees with human labels only 68.1% of the time and is markedly weaker on sentiment. We did not run a native-speaker review of the generated text itself, so nothing in this paper is evidence about the cultural, dialectal, or orthographic adequacy of our synthetic examples. Automatic channels can miss exactly those failures, and a selector that looks clean on our channels could still retain text a speaker would reject.

Leakage and shortcut detectors. Our exact n-gram, shingle-Jaccard, and LaBSE-based leakage checks target observable contamination and cannot certify the absence of memorisation in the generator's training data. The shortcut detector uses a fixed feature inventory, and no candidate reaches τ_H=0.70 (maximum H is 0.489), so the shortcut gate never fires, though H still contributes to ranking. All selectors read the same audit fields, but because they weight those fields differently, their rankings may respond differently to detector error.

Score-weight calibration. We chose the weights in S_i on a held-out Setswana development split to equalise per-channel contributions, and we ship them in `configs/default.yaml` so others can vary them without regenerating candidates. We did not release the calibration split or its script, so τ_L and the α weights should be read as fixed protocol choices rather than a derivation a reader can reproduce. A $\pm 10\%$ sensitivity sweep on τ and α would quantify how far the V2 pool moves under nearby protocol choices. We did not run it.

Baseline coverage. AlpaGasus and DEITA are budget-matched reimplementations of each method's *selection criterion* rather than runs of the original codebases, all reading the shared audit record so the comparison isolates the selection rule. Our numbers therefore characterise the criteria and not the published systems. The U+D control computes no gradient influence and is named for what it computes. We additionally implement LESS itself (§4.4), on one seed and one training loop that is not the main replay's, so its absolute values are not comparable to Table 1.

Scope of the ablation. §5.1 measures what the audit's strict gate costs a strong selector, on eight cells at three seeds. It is not a clean causal decomposition of CoSDA-V2, which applies no gate at all, and in the two underfilled cells it varies training-set size alongside the selection rule, so filtering and data volume are confounded exactly where the effect is largest. It does not establish which channel inside the gate does the pruning, and it does not test the budget-aware repair it suggests. The diversity and judge-resolution interpretations remain trace-grounded rather than ablated mechanisms, though the latter now has a multi-judge sensitivity check.

Ethical Considerations

Synthetic data for low-resource communities can ease annotation pressure, but it can also distort cultural context, normalise incorrect language use, or surface private information from web-trained generators (Bender et al., 2021; Carlini et al., 2021). We designed CoSDA to expose a limited subset of these risks before selection, logging provenance, checking leakage at three levels, and measuring counterfactual consistency before an example enters a retained pool. These automatic channels do not assess cultural or dialectal adequacy, which still requires review by speakers of the target language.

Our replay uses MasakhaNEWS (Adelani et al., 2023) and AfriSenti (Muhammad et al., 2023), whose licence metadata are internally inconsistent: the MasakhaNEWS card declares `afl-3.0` while its prose states CC 4.0 Non-Commercial, and the AfriSenti card declares `cc-by-nc-sa-2.0` while its prose states CC BY 4.0.[1] Both corpora also carry third-party text, BBC and VOA article bodies in MasakhaNEWS and tweet text in AfriSenti. Because we did not establish that these licences cover redistribution of every underlying text, we withhold source text and ship record identifiers and digests instead. Every released record carries its source dataset identifier, its seed provenance, and a `generator` field, so generated text is marked as generated. Anyone who identifies content they did not consent to release can request removal through the artifact repository's issue tracker. Practice in this area recommends accompanying data statements and datasheets (Bender and Friedman, 2018; Gebru et al., 2021), which we regard as the right next step for this release.

We report automatic audit metrics rather than a human annotation study, so we claim no human-validated cultural plausibility. We did validate our judge against human-annotated benchmark labels (§4.4) and found it separates correct from flipped labels while agreeing with human labels only 68.1% of the time, and less reliably on sentiment than on news topic. No native speaker reviewed the generated text itself. Readers should therefore treat every audit-quality number here as a machine measurement, and anyone deploying selected examples should add community review by speakers of the

[1] Dataset cards consulted 2026-08-23: https://huggingface.co/datasets/masakhane/masakhanews and https://huggingface.co/datasets/masakhane/afrisenti.

target language, especially for political news and sentiment data.

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A Reproducibility Checklist

How to read the appendix. We organise the appendix as a trace from evidence to claim. This section names the source artifacts and replay contract. Appendix B describes the prompts and audit fields that create the per-candidate records. Appendix C gives the full detail tables and appendix plot. The main paper reports the claim-carrying summaries, and the appendix exposes the rows and artifacts behind them.

Canonical replay outputs. The main replay is a single per-cell table indexed by task, language, and selector. Each row records the retained-set size, judged label correctness, judged quality, counterfactual consistency C, leakage score L, shortcut score H, hard-reject rate, and downstream Macro-F1 from the deterministic Hugging Face classification path. That table populates the main-replay tables and figures. The extended replay of §4.4 adds the separate result files listed in the artifact README, and the per-candidate statistics we report, namely the diversity column of Table 10 and the C_i pass rates of §2, are recomputed from the released audit records under `runs/select1_seed13/`. Aggregates (selector-level means, Spearman correlations, bootstrap CI, leave-one-cell-out, and non-degenerate-subset statistics) are deterministic functions of these files, not separate experiments.

Authoritative artifact paths. The artifact's `README.md` lists the files treated as authoritative, and `results/main_replay/claim_ledger.csv` maps the audit channels, extended-replay means, AfroXLMR means and ablation values to their source file and computation.

Released artifacts. We release four artifact groups. The dataset manifest covers the dataset id, split paths, split sizes, and label counts for MasakhaNEWS and AfriSenti. The candidate group holds the generated examples and their per-candidate audit records, namely utility components, leakage components, shortcut features, diversity, counterfactual consistency, judged label correctness, judged quality, and the hard-reject decision. The replay group holds the per-cell table described above for all selectors plus the gold-only reference. The script group downloads data, generates candidates, scores audit channels, applies each selector, fine-tunes the downstream classifier, and emits the replay table. The per-candidate audit records and the seven per-selector retained pools for all eight cells ship under `runs/select1_seed13/`. The extended replay of §4.4 ships as a separate result group with its own run scripts and per-run JSON, kept apart from the main replay because it was fitted on different hardware. The release separates candidate-level audit records from selected-set summaries so that a reader can recompute a selector without trusting the aggregate tables.

Claim ledger. The released claim ledger (`results/main_replay/claim_ledger.csv`) maps the audit channels, the extended-replay means, the AfroXLMR means and the ablation values to the result file and computation that produces each, and `scripts/make_claim_ledger.py` regenerates it. A single command, `reproduce.sh`, recomputes the extended-replay aggregates and the main-replay statistics of §4.3 from the committed per-run results using only the Python standard library, and the released `ANALYSIS_OUTPUT.txt` and `judge_agreement_output.txt` are the reference outputs to diff against.

Default hyperparameters. For the audit channels, α_U=1.0, α_C=0.75, α_D=0.50, α_L=1.0, and α_H=0.75; hard-reject thresholds are τ_L=0.15, τ_C=0.60, and τ_H=0.70. These values were fixed before the replay and reused unchanged across all eight task-language cells. Downstream fine-tuning uses the deterministic Hugging Face classification path (`cosda/hf_training.py`): 3 epochs, batch size 16, learning rate 2×10^{-5}, maximum length 256, warmup ratio 0.1, evaluating on dev each epoch and restoring the best dev Macro-F1 checkpoint. The main-replay seed is 13 and the gold budget is b=64 with synthetic multiplier m=3.

Calibration of thresholds and weights. The thresholds and weights are protocol choices fixed before the replay (see the end of §2): τ_L=0.15 bounds the composite leakage risk L_i rather than a raw cosine, τ_H=0.70 bounds the shortcut score H_i, τ_C=0.60 gates $C_i \in [0, 1]$, and the score weights $(\alpha_U, \alpha_C, \alpha_D, \alpha_L, \alpha_H)$ equalise per-channel contributions in $[0, 1]$ on a held-out Setswana split we do not release. All four groups are identical across the eight reported cells and remain fixed throughout the replay.

Extended-replay protocol. The extension reuses the seed-13 retained pools verbatim, so audit channels are unchanged and only the downstream fit is recomputed: seeds 13/21/42 with XLM-RoBERTa base, and seed 13 with AfroXLMR base (Alabi et al., 2022) (`Davlan/afro-xlmr-base`), same deterministic training path and hyperparameters throughout. The two additional judges are Qwen2.5-14B-Instruct and Phi-3.5-mini-instruct (Abdin et al., 2024), run with the same judging prompt as the main judge.

AI assistance. Beyond the language models that are the objects of study in this paper (§2, §3), we used AI assistants in two ways: code assistance during implementation, and language polishing during manuscript preparation. Every reported number was recomputed from the released result files, and the authors take full responsibility for all claims and interpretations.

Deferred experiments. Native-speaker review of the generated text, generator replacement, multi-prompt judge sweeps, budget sweeps, per-channel decomposition of the hard-reject gate, the budget-aware hard-reject repair proposed in §5.1, and task extensions to NER, intent/slot filling, and summarisation are not claimed as completed results in this paper.

B Prompts and Audit Protocol

Purpose. We use the prompt and audit protocol to make synthetic examples inspectable before selection. The generator creates candidate examples and counterfactual edits, and the audit record then stores both judge-facing quality fields and non-judge diagnostic fields. The downstream classifier never sees the audit fields directly. It sees only the retained synthetic examples chosen by each selector.

Generation prompt template. The generation prompt fixes task, language, label schema, seed examples, and output format, and includes the forbidden-behaviour clause. The template as shipped in `cosda/generation.py` is reproduced below. The prompt hash stored in each candidate record is used for traceability, not as a modelling feature.

```
Task: {task}
Language: {language}
Allowed labels: {labels}
Seed label: {label}
Seed text excerpt:
{seed_excerpt}
Generate {n} new examples in the same language and label.
Each generated text should be a concise news item of 60-120
words. [For sentiment cells this line instead reads: Each
generated text should be concise and natural, usually under
60 words.] Do not copy the seed text or held-out text. Do not
translate an English template. Return valid JSON only, with
no Markdown fences or commentary. Return JSON list: [{"text":
str, "label": str, "confidence": float}].
```

Counterfactual edit template. For each candidate x_i, the generator receives x_i and produces x'_i by changing one classification-relevant variable, such as the topic or sentiment label cue, while preserving language, register, and as much non-label content as possible. The expected flipped label is stored with the counterfactual pair and used by the counterfactual-consistency score. A pair is therefore evaluated as a data-quality check: the candidate should be fluent and labelled correctly, and the paired edit should move the label in the expected direction.

Automatic audit fields. The replay records judged label correctness, judged quality, counterfactual consistency (C), leakage score (L), shortcut score (H), hard-reject status, and selected-set membership for each retained-set baseline. These fields support Tables 1 and 10. They are automatic audit signals, not human annotation outcomes.

Selector membership fields. Each selector receives the same candidate pool within a task-language cell. We release one retained-pool file per selector next to the audit record, covering all seven selectors, rather than recording membership as a field inside each candidate. This shared-pool design is why the appendix tables can compare audit quality and downstream Macro-F1 without confounding the result with different generation runs.

Hard-reject interpretation. The interpretation of the hard-reject column, an audit decision over each selector's retained pool rather than a ground-truth label, is described in §2. The downstream results separate this retained-pool property from Macro-F1.

Qualitative examples. Table 7 gives four candidates drawn from the per-candidate audit records, illustrating one *kept* candidate, one rejected for counterfactual consistency, one rejected for leakage, and the highest shortcut score in the replay, which the gate does not reject. The candidates are shown with a brief description of the trigger, not the raw text, since text in four African languages is hard to read inline. The release ships the full audit record for every candidate, including the raw text, the generator's counterfactual edit, the per-channel scores, and the hard-reject reason.

Future human validation. We did not conduct speaker validation. A future study should ask L1 or strong L2 speakers to judge label correctness, fluency, cultural plausibility, leakage suspicion, and counterfactual validity.

C Detail Tables and Plots

This appendix contains the detail evidence behind the compact main-body results. Numbers and plots in this section come from the per-cell replay table described in Appendix A, except the diversity column and the AfroXLMR results, which are recomputed from the released audit records and the extended-replay files. We include these tables so readers can trace the main audit, downstream, and per-cell results to the same 8-cell replay.

Benchmark composition

The completed replay is deliberately narrow: two classification datasets and four languages. This scope keeps counterfactual validation task-specific

Status	Trigger and audit profile
Kept	Swahili business candidate (`5328207e1efef02d`); LC= 1.0, C=0.72, L=0.00, H=0.02. Passes all hard-reject thresholds and is ranked in the top-64 by every quality-aware selector.
Reject (CF)	Amharic sentiment candidate (`eabcd5ba208b51da`) where the judge scores label correctness 0 and counterfactual validity v_{cf}=0.7; C_i=0.59 falls below τ_C=0.6 and the candidate is hard-rejected on the counterfactual channel.
Reject (leakage)	Yoruba entertainment candidate (`5554f5d6339bd4c8`) with a 10-gram match against the MasakhaNEWS dev/test union (E_i=L_i=1.0); the leakage gate fires before the candidate reaches the reranker. It is the only leakage rejection in the eight cells.
No shortcut reject	The largest shortcut score in the replay is a Yoruba sentiment candidate (`96be681d80e672a9`) at H=0.49, from a label-word hit plus atypical length. It stays below τ_H=0.70, so the shortcut gate fires on no candidate in the eight cells.

Table 7: Four candidates: kept, counterfactual-rejected, leakage-rejected, and the highest shortcut score in the replay. Values are taken from the released audit records.

Dataset	Task	Languages
MasakhaNEWS (Adelani et al., 2023)	News topic classification	amh, hau, swa, yor
AfriSenti (Muhammad et al., 2023)	Sentiment classification	amh, hau, swa, yor

Table 8: Our completed 8-cell evaluation, with Macro-F1 as the metric in every cell. Broader NER, intent, and summarisation settings need validators beyond our classification protocol.

and avoids mixing completed classification evidence with planned extensions.

Full downstream table with gold-only

Table 9 gives the downstream view used to compute the aggregate Macro-F1 claims. The Gold-only row is a reference point for low-resource fine-tuning without synthetic examples. It is not included in the audit-vs-downstream Spearman calculations because it has no retained synthetic pool to audit.

Full per-selector audit table

Table 10 gives the full audit view over selected synthetic examples. Our main text focuses on LC, Q, hard-reject, C, and H because those channels tie most directly to the gap between audit quality and utility, and the full table adds leakage L, where all means are small but selector differences stay visible.

Baseline	n	F1↑	Std	Wins	Δ
Gold only	8	0.116	0.074	2/8	−0.051
Naive	8	0.167	0.068	n/a	n/a
AlpaGasus	8	**0.202**	0.099	4/8	+0.036
DEITA	8	0.128	0.071	4/8	−0.039
CoSDA-EQ	8	0.133	0.109	1/8	−0.034
CoSDA-V2	8	0.163	0.103	2/8	−0.004

Table 9: Full 8-cell replay including the Gold-only reference, with wins and Δ measured against naive. CoSDA-V2 sits near naive on the mean, and §4.3 places that difference inside cell-level uncertainty.

Sel.	LC↑	Q↑	C ↑	D ↑	L ↓	H ↓	Rej.↓
Naive	0.767	0.662	0.587	0.599	0.003	0.066	0.486
AlpaGasus	0.890	**0.749**	0.653	0.590	0.006	0.065	0.246
DEITA	0.869	0.737	0.637	**0.649**	0.005	0.070	0.299
CoSDA-EQ	0.836	0.718	**0.686**	0.604	**0.002**	0.058	**0.109**
CoSDA-V2	**0.904**	0.746	0.674	0.602	0.003	**0.057**	0.162

Table 10: Full judge and audit quality of selected synthetic data, including the leakage column L omitted from the main body. Every quality-aware selector improves on naive across LC, Q, C, and hard-reject.

The diversity column requires explanation. It is the mean per-candidate D_i over each selector's 64-example retained pool, which we compute by joining per-candidate audit records with that selector's selected JSONL. That column weighs against variety preservation as the source of AlpaGasus's downstream lead in §5, since AlpaGasus scores D=0.590, lower than every CoSDA variant, while DEITA, which explicitly weights diversity, records the highest D at 0.649 and still finishes last downstream.

Per-cell Macro-F1 (main replay)

Table 11 gives the per-cell downstream view behind §4.3, for all five main-replay selectors with each selector's delta against naive.

Extended-replay detail

This subsection holds the row-level evidence behind §4.4. It is computed on the same seed-13 retained pools as the main replay, so the audit columns are unchanged and only the downstream fit is recomputed. The AfroXLMR gains quoted in Table 12 are differences against this pipeline's own seed-13 XLM-R fit, which the released `ANALYSIS_OUTPUT.txt` prints, and not against the main-replay XLM-R column of Table 11; §3 gives the size of that pipeline difference.

Leave-one-cell-out and per-cell deltas

Table 13 and Figure 5 support §4.3. Our eight-cell V2-minus-naive mean of −0.004 is built from

Cell	Naive	AG	DEITA	EQ	V2	Δ
news/amh	0.286	0.342	0.105	0.105	0.105	−0.181
news/hau	0.102	0.195	0.039	0.039	0.066	−0.036
news/swa	0.069	0.184	0.109	0.012	0.198	+0.129
news/yor	0.128	0.065	0.065	0.065	0.065	−0.062
sent./amh	0.154	0.067	0.067	0.067	0.069	−0.085
sent./hau	0.168	0.339	0.196	0.349	0.375	+0.206
sent./swa	0.248	0.248	0.253	0.248	0.248	+0.000
sent./yor	0.178	0.176	0.185	0.176	0.176	−0.002
mean	0.167	**0.202**	0.128	0.133	0.163	−0.004

Table 11: Per-cell Macro-F1 in the main replay (seed 13, XLM-R). AG is AlpaGasus; EQ and V2 are the CoSDA variants; Δ is V2 minus naive. The Gold-only reference is in Appendix Table 9.

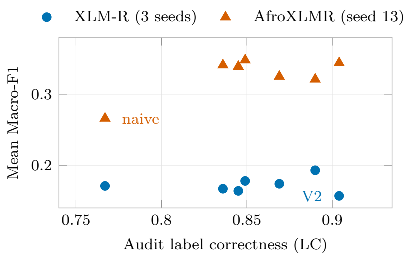

Figure 4: Audit LC against mean Macro-F1 for all seven selectors. XLM-R shows no upward trend, while AfroXLMR raises every selector's Macro-F1 and increases alignment between the two rankings.

cell effects of opposite sign, and removing Hausa sentiment, the single large positive cell, moves it to −0.034. We include this view because an eight-cell mean can be dominated by one cell without that being visible in the aggregate. Restricting instead to the five cells that are not near-degenerate gives selector means of 0.226 for AlpaGasus, 0.163 for CoSDA-V2, 0.156 for naive, 0.114 for CoSDA-EQ, 0.103 for DEITA, and 0.087 for Gold-only.

Within-cell Spearman ranks

Figure 2 in the main body is the canonical presentation of our primary statistic, with four cells

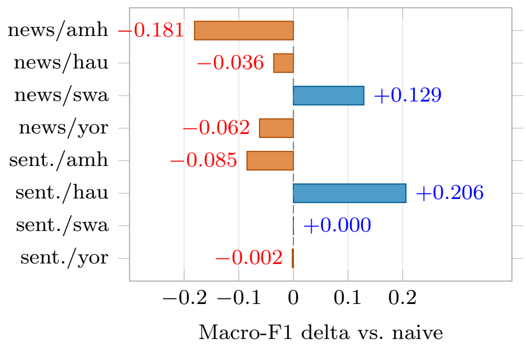

Figure 5: Per-cell Macro-F1 delta of CoSDA-V2 against naive. Each bar is one task-language cell, and positive values mean CoSDA-V2 wins.

Cell	Naive	AG	DEITA	U+D	AG+HR	EQ	V2
news/amh	0.591	0.285	0.607	0.376	0.443	0.499	0.612
news/hau	0.205	0.180	0.244	0.314	0.262	0.273	0.291
news/swa	0.311	0.218	0.247	0.303	0.232	0.374	0.345
news/yor	0.238	0.455	0.369	0.437	0.380	0.311	0.304
sent./amh	0.162	0.364	0.191	0.443	0.329	0.292	0.191
sent./hau	0.257	0.361	0.459	0.378	0.347	0.411	0.350
sent./swa	0.242	0.362	0.152	0.268	0.374	0.249	0.318
sent./yor	0.120	0.347	0.328	0.263	0.344	0.318	0.338
mean	0.266	0.321	0.325	**0.348**	0.339	0.341	0.344

Table 12: Per-cell Macro-F1 with the AfroXLMR backbone (Alabi et al., 2022) at seed 13, on the same retained pools. Every selector gains over the extended pipeline's own seed-13 XLM-R fit, from +0.118 for naive to +0.182 for AlpaGasus+HR.

Cell removed	Mean Δ (V2−Naive)↑
(all 8 cells)	−0.004
news/amh	+0.022
news/hau	+0.001
news/swa	−0.023
news/yor	+0.005
sent./amh	+0.008
sentiment/hau	**−0.034**
sent./swa	−0.004
sent./yor	−0.004

Table 13: Leave-one-cell-out mean Δ of CoSDA-V2 vs. naive. The first row is the all-cell baseline, and subsequent rows show the mean when the named cell is excluded.

positive, four negative, a mean of +0.04, and a median of 0.00. Its values come from the main-replay CSV named in the artifact README, and §4.4 recomputes the statistic over seven selectors in the extended pipeline, where the seed-13 value is +0.006 rather than +0.04.